GCSE Biology
Exam Walkthrough Pack

Step-by-step worked exam-style questions with full explanations

Original JDScience exam-style questions — not copied from past papers

Introduction

This Biology Exam Walkthrough Pack is designed to help you understand how to approach GCSE and IGCSE exam-style questions. Every question in this booklet is an original JDScience item written to match common specification skills — it is not copied from AQA, Edexcel, OCR or other past papers.

How to use this pack: read each question, cover the walkthrough, and try answering in exam conditions first. Then read what the question is asking, follow the step-by-step thinking, compare your answer with the model answer, and study the mark breakdown to see exactly how marks are awarded.

Command words tell you what type of answer is required. Section 1 explains the most common GCSE science command words. Always underline the command word before you start writing.

Showing working matters: in calculations, examiners award method marks even when the final answer is wrong. In longer answers, a clear logical sequence helps you secure every available mark.

Examiners award marks for correct science linked to the question. Keywords from the specification matter, but a correct idea in your own words can still score. Extended questions often use level-based marking: plan before you write, use paragraphs, and link ideas.

To move from Grade 4/5 to Grade 7/8/9: use precise terminology, explain mechanisms (not just names), include data or examples where asked, and check units and significant figures in calculations. For six-mark questions, a short plan and linked paragraphs beat a long unordered list.

Section 1: Command word guide

State

What you must do: Give a short, factual answer — often one word, number or phrase. No explanation is required unless the question also asks you to explain.

Common mistake: Writing a long paragraph when a single fact is enough, or giving an explanation when only a fact was asked for.

Model sentence starter: The [quantity/name] is …

Give

What you must do: Provide the information requested. Similar to state, but may need a short phrase or named example.

Common mistake: Listing unrelated facts that do not answer the question.

Model sentence starter: One example is … / The value is …

Describe

What you must do: Say what happens or what something is like. Focus on observations, patterns or features. You do not need to say why.

Common mistake: Explaining causes when the command word is only describe.

Model sentence starter: First … then … / The pattern shows …

Explain

What you must do: Give reasons why something happens. Link cause and effect using science ideas and key terms.

Common mistake: Describing what happens without saying why, or using everyday language instead of science.

Model sentence starter: This happens because … which means …

Calculate

What you must do: Work out a numerical answer. Show your working, include units, and give the final value to the required precision.

Common mistake: Missing units, no working, or using the wrong equation.

Model sentence starter: Use … = … / Substitute: …

Compare

What you must do: Identify similarities and/or differences between two or more things. Refer to both sides.

Common mistake: Only describing one item, or listing features without comparing.

Model sentence starter: Both … however … / X has … whereas Y …

Evaluate

What you must do: Weigh up evidence or options, consider strengths and weaknesses, and reach a supported judgement.

Common mistake: Only listing advantages with no conclusion or balanced comment.

Model sentence starter: An advantage is … A limitation is … Overall …

Suggest

What you must do: Apply your knowledge to a new situation. Your answer should be sensible and scientifically plausible.

Common mistake: Repeating textbook facts that do not fit the context given.

Model sentence starter: A possible reason is … / This could be because …

Justify

What you must do: Give evidence or reasons that support a choice, conclusion or statement already made.

Common mistake: Restating the claim without giving supporting science.

Model sentence starter: This is supported by … because …

Determine

What you must do: Find out a value or conclusion using data, a graph or a calculation. Show how you reached your answer.

Common mistake: Reading a graph incorrectly or not showing how the value was obtained.

Model sentence starter: From the graph … / Using the data …

Analyse

What you must do: Break information into parts, interpret patterns or trends, and explain what the data shows.

Common mistake: Describing data without interpreting it, or ignoring anomalies.

Model sentence starter: The data shows that … / As X increases, Y …

Section 2: Cell biology

QUESTION 1

Name the organelle that controls the activities of the cell.

QUESTION TYPE

Topic: Cell biology | **Command word:** state | **Marks:** 1 | **Skill tested:** Cell structure

WHAT THE QUESTION IS ASKING

Identify the control centre of the cell.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The nucleus.

MARK BREAKDOWN

Mark 1: The nucleus.

COMMON MISTAKES

- Writing mitochondrion.
- Giving a description instead of a name.

EXAMINER TIP

For one-mark recall, give a single precise term.

QUESTION 2

State the function of ribosomes.

QUESTION TYPE

Topic: Cell biology | **Command word:** state | **Marks:** 1 | **Skill tested:** Cell structure

WHAT THE QUESTION IS ASKING

State what ribosomes do.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Protein synthesis.

MARK BREAKDOWN

Mark 1: Protein synthesis.

COMMON MISTAKES

- Saying respiration.
- Saying they make DNA.

EXAMINER TIP

Link ribosomes to proteins, not energy.

QUESTION 3

Give the name of the structure that makes a plant cell rigid.

QUESTION TYPE

Topic: Cell biology | **Command word:** give | **Marks:** 1 | **Skill tested:** Plant cells

WHAT THE QUESTION IS ASKING

Name the plant cell wall component.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Cell wall (made of cellulose).

MARK BREAKDOWN

Mark 1: Cell wall (made of cellulose).

COMMON MISTAKES

- Saying membrane.
- Saying vacuole.

EXAMINER TIP

Cell wall is outside the membrane in plant cells.

QUESTION 4

Describe two differences between a prokaryotic cell and a eukaryotic cell.

QUESTION TYPE

Topic: Cell biology | **Command word:** describe | **Marks:** 2 | **Skill tested:** Cell types

WHAT THE QUESTION IS ASKING

Compare prokaryotes and eukaryotes structurally.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Prokaryotic cells have no nucleus — DNA is free in the cytoplasm — and are smaller. Eukaryotic cells have a membrane-bound nucleus and membrane-bound organelles.

MARK BREAKDOWN

Mark 1: One valid structural difference.

Mark 2: Second valid structural difference.

COMMON MISTAKES

- Only describing one type.
- Stating functions without structural differences.

EXAMINER TIP

Nucleus presence and size are reliable two-mark pairs.

QUESTION 5

Explain why root hair cells have many mitochondria.

QUESTION TYPE

Topic: Cell biology | **Command word:** explain | **Marks:** 2 | **Skill tested:** Specialised cells

WHAT THE QUESTION IS ASKING

Link structure to function for active transport.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Root hair cells absorb mineral ions by active transport, which requires ATP from aerobic respiration in mitochondria.

MARK BREAKDOWN

Mark 1: Active transport or ion uptake mentioned.

Mark 2: Link to ATP or respiration in mitochondria.

COMMON MISTAKES

• Saying photosynthesis.

• Not linking mitochondria to energy need.

EXAMINER TIP

Always connect organelle to the process that needs energy.

QUESTION 6

Compare diffusion and osmosis.

QUESTION TYPE

Topic: Cell biology | **Command word:** compare | **Marks:** 2 | **Skill tested:** Transport

WHAT THE QUESTION IS ASKING

State a similarity and a difference.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

Both involve passive movement down a concentration gradient without energy. Diffusion can involve any particle; osmosis is the movement of water only across a partially permeable membrane.

MARK BREAKDOWN

Mark 1: Valid similarity.

Mark 2: Valid difference about water or membrane.

COMMON MISTAKES

- Saying osmosis needs energy.
- Not mentioning water specifically.

EXAMINER TIP

Use 'both … however …' structure.

QUESTION 7

Explain how the structure of the cell-surface membrane helps it control what enters and leaves the cell.

QUESTION TYPE

Topic: Cell biology | **Command word:** explain | **Marks:** 3 | **Skill tested:** Membranes

WHAT THE QUESTION IS ASKING

Link phospholipid bilayer and proteins to selective permeability.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The membrane is a phospholipid bilayer forming a barrier to many polar molecules. Embedded proteins act as channels or carriers for specific substances, so only selected ions and molecules can cross.

MARK BREAKDOWN

Mark 1: Phospholipid bilayer or barrier idea.

Mark 2: Proteins as channels or carriers.

Mark 3: Selective control of substances.

COMMON MISTAKES

- Calling the membrane fully permeable.
- Ignoring protein role.

EXAMINER TIP

Mention both the lipid barrier and protein channels.

QUESTION 8

Describe the stages of mitosis in order.

QUESTION TYPE

Topic: Cell biology | **Command word:** describe | **Marks:** 3 | **Skill tested:** Cell division

WHAT THE QUESTION IS ASKING

Outline the sequence of mitosis.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER